

Fault Code: 551

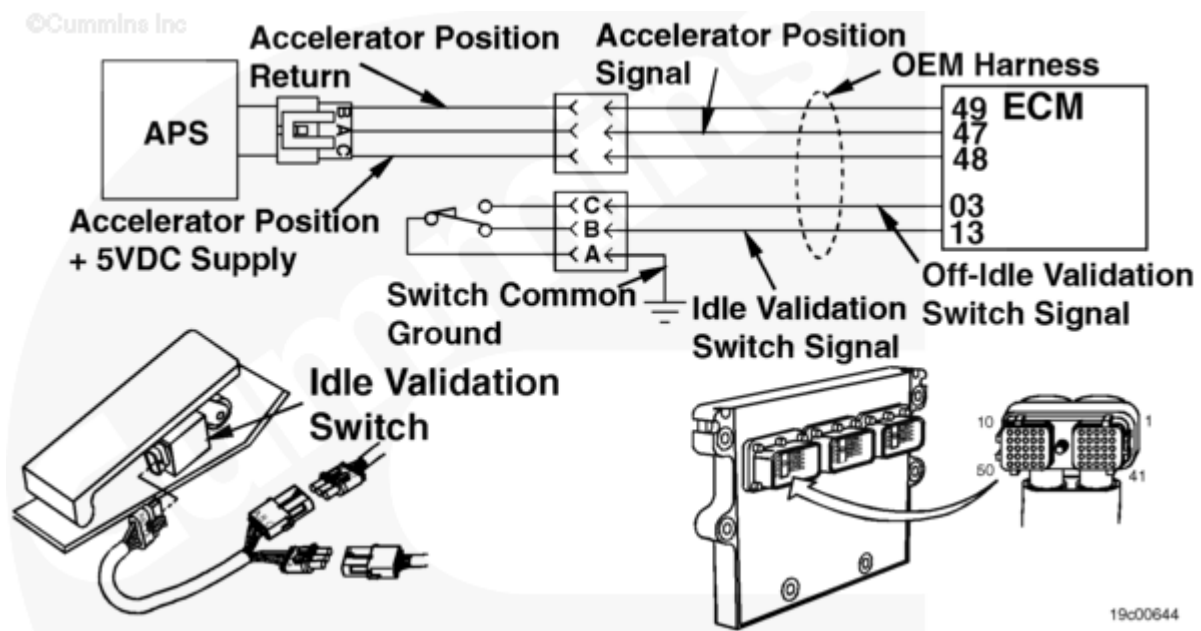
SSS

Idle Validation Switch Circuit

 Printable Version

Overview

Codes	Reason	Effect
Fault Code: 551 PID(P): S230 SPN: 558 FMI: 4/4 Lamp: Yellow SRT:	No voltage detected simultaneously on both the idle validation off-idle and on-idle circuits.	Engine will only idle.



Idle Validation Switch Circuit

Circuit Description

The idle validation switch (IVS) is used by the electronic control module (ECM) to indicate when the accelerator pedal is released (on-idle) or depressed (off-idle). The switch is adjusted at the factory to switch from on-idle to off-idle at the correct accelerator pedal position. Solid-state integrated sensor switch (SSS) is an active electronic component that requires voltage to inspect the circuit.

Component Location

The SSS is located on the accelerator pedal assembly.

Shoptalk

This fault code is usually caused by a loose connection, an accelerator pedal that is **not** calibrated, or an IVS that is wired wrong. Breakout cable, Part Number 3824892, is required to check the voltage levels of the SSS.

- If all wiring and sensor checks are good, replace the idle validation switch circuit wires, between the accelerator pedal and ECM, with new wires. Run the wires through or around the bulkhead without using the bulkhead connector. Test the truck with the test wires in place. If the fault clears, replace the OEM harness. Seal the openings in the bulkhead around the connector and wires to prevent toxic and noxious fumes from seeping into the cab.

Note: Check that the three idle validation switch circuit wires are twisted together.

- The ECM and accelerator pedal assembly **must** be electrically calibrated to each other for proper engine response. They **must** be calibrated when an accelerator pedal is initially installed, replaced, when an ECM is replaced, when a new calibration is downloaded to the ECM, and when the accelerator pedal wiring is disconnected while the vehicle keyswitch is on.
- With keyswitch on, gradually push the accelerator to the floor and release. Depress and release the accelerator pedal three times. This maneuver will recalibrate the ECM and accelerator pedal.

Refer to Troubleshooting Fault Code t05-551sss (</qs3/pubsys2/xml/en/procedures/82/82-t05-551sss.html>)